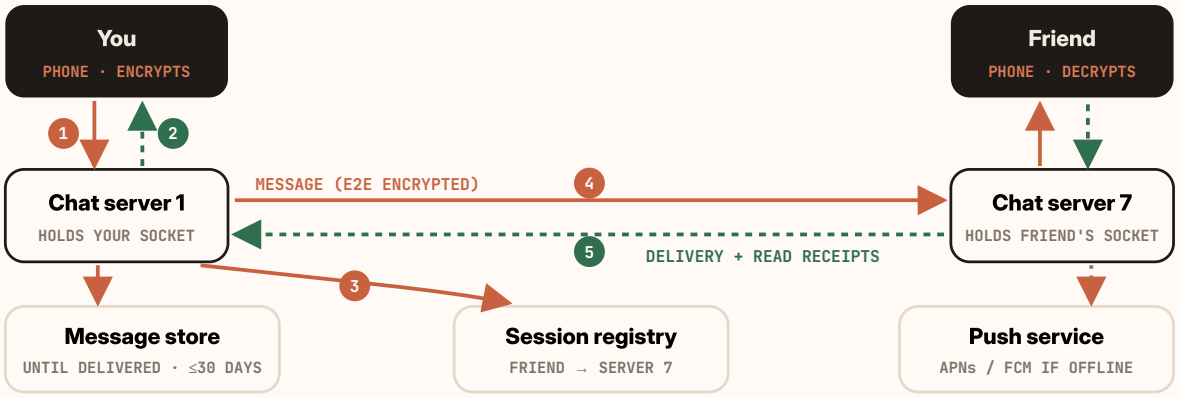
• SYSTEM DESIGN UNLOCKED · EP 03

Design WhatsApp's *Blue Ticks*

How a message travels, and how ✓, ✓✓ and blue find their way back to you.

THE FLOW



1 Phone encrypts, sends over its one open connection
2 Server stores it, ACKs back → ✓
3 Server looks up friend in the session registry
4 Forwards to server 7 → pushed down friend's socket
5 Delivery receipt back, stored copy deleted → ✓✓
6 Friend opens chat → read receipt → ✓✓ blue

WHAT EACH TICK MEANS (BACKEND)

✓ **Sent.** The server stored the message and ACKed your phone. Your friend doesn't have it yet.

✓✓ **Delivered.** Friend's phone (or a linked device) sent a delivery receipt. Server deletes its copy.

✓✓ **Read.** Friend opened the chat; a read receipt travelled back the same path.

REQUIREMENTS

- Real-time delivery, low latency
- Works when the receiver is offline
- No lost messages, no duplicates
- End-to-end encrypted: server can't read
- Status (✓ / ✓✓ / blue) for every message

~2M
CONNECTIONS / SERVER (ERLANG, 2012-14)

30 days
MAX HOLD FOR UNDELIVERED

1
OPEN CONNECTION PER DEVICE

EDGE CASES → FIX

Phone off → message waits in the store (≤30 days) + push notification. Sender stays on ✓.

ACK lost → phone retries with the *same message ID*; server drops the duplicate (idempotency).

Read receipts off → stays grey in 1:1 chats (and you stop seeing others' blue too).

Groups → ✓✓ when everyone received, blue only when everyone read.

Linked devices → delivered once any of the friend's devices gets it.

TRADE-OFFS TO SAY OUT LOUD

Push over polling: instant + battery-friendly, but you must hold millions of open sockets.

Store until delivered: private + cheap storage, but no server-side history (backups live on the device).

At-least-once + dedupe: never lose a message; message IDs stop doubles.

Receipts are messages too: ~2 extra tiny messages per message = more traffic.

INTERVIEW KEYWORDS TO SAY

`persistent connection` `WebSocket / TCP` `ACK` `delivery receipt` `read receipt` `message ID` `idempotency` `at-least-once delivery` `session registry` `store-and-forward` `push (APNs / FCM)` `end-to-end encryption` `Signal Protocol`

Sources: faq.whatsapp.com/665923838265756 · whatsapp.com/legal/privacy-policy · whatsapp.com/security/WhatsApp-Security-Whitepaper.pdf · highscalability.com (WhatsApp architecture). Message-ID dedupe is the standard interview design, not a claim about WhatsApp's code.